SITYOG INSTITUTE OF TECHNOLOGY

Growth Centre, Jasoiya More, Aurangabad, Bihar – 824101

DEPARTMENT OF COMPUTER APPLICATION

Report on

STUDENT TRAINING MANAGEMENT SYSTEM

This report is submitted in partial fulfilment of the requirements for the award of the degree

Bachelor of Computer Application

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DEPARTMENT OF COMPUTER APPLICATION

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DECLARATION CERTIFICATE

This is to certify the work presented in the Project Report in "**Student Training Management System**", in partial fulfilment of the requirement for the award of Degree of Bachelor of Computer Application of SITYOG Institute of Technology, Aurangabad, is an authentic work carried out by us under the supervision of **Mr. S.K Ojha**, Assistant Professor, Department of Computer Applications.

To the best of our knowledge, the content of this report does not form a basis for the award of any previous Degree to anyone else.

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CERTIFICATE OF APPROVAL

The foregoing Project REPORT on "**Student Training Management System**" is hereby approved as a creditable study of Project report and has been presented in satisfactory manner to warrant its acceptance as prerequisite to the degree for which it has been submitted.

It is understood that by this undersigned do not necessarily endorse any conclusion drawn or opinion expressed therein, but approve the report for the purpose for which it is submitted.

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Chapter 1

Introduction

1.1 Background

The rapid advancement of information technology has transformed the way educational institutions manage their administrative and academic functions. Traditionally, student records, training programmes, and academic results were maintained manually through paper registers and spreadsheets — methods that are time-consuming, error-prone, and difficult to retrieve quickly.

In the context of technical and professional education, managing training and internship records is particularly challenging. Students are placed with various companies as part of their curriculum, and keeping track of each student's training status, project, company, and performance requires a well-organised system.

The **Student Training Management System** is developed to address these challenges by providing a centralised, web-based platform for managing student personal details, company information, training records, and academic results.

1.2 Problem Statement

The following problems were identified at educational institutions like Sityog Institute of Technology:

- Manual record-keeping is slow and error-prone; data retrieval is tedious.
- No centralised system exists to track which students are training with which companies.
- Academic results are stored separately without link to training records.
- Generating consolidated reports is laborious and time-consuming.
- No secure authentication to prevent unauthorised access to student data.

1.3 Objectives

- **Develop a web-based system** for managing student details, training records, company information, and academic results.
- **Implement secure authentication** so only authorised administrators can access and modify data.
- **Provide real-time search and filtering** for quick access to records by name, roll number, or other parameters.
- **Maintain training and internship records** including company, project, duration, and status.
- **Track academic results** across semesters and generate result summaries.
- **Generate comprehensive reports** covering all modules in a consolidated view.
- **Build a modern, responsive UI** that is intuitive for administrative staff.

1.4 Scope of the Project

In Scope:

- Management of student personal and academic information.
- Management of company and placement records.
- Tracking of student training and internship programmes.
- Recording and viewing of semester-wise academic results.
- Consolidated report generation for all modules.
- Secure admin login and JWT session management.
- Individual student profile view with complete history.

Out of Scope:

- Student-facing portal (students cannot log in in this version).
- Fee or payment management.
- Attendance management.
- Email or SMS notification system.
- Multi-campus or multi-department support.

Chapter 2

Literature Review

2.1 Existing Systems

2.1.1 Traditional Manual Systems

Many small and medium-sized educational institutions in India still rely on paper registers and files. While simple, these methods require significant storage space, make retrieval tedious, and are highly susceptible to data loss due to damage or misplacement.

2.1.2 Microsoft Excel-Based Systems

Spreadsheets offer an improvement over paper-based records but lack multi-user access, role-based security, and the ability to generate dynamic cross-module reports. They also have no validation and are prone to accidental modification.

2.1.3 Enterprise ERP Systems

Large universities use enterprise ERP systems such as SAP or Oracle. While comprehensive, these are prohibitively expensive, require dedicated IT infrastructure, and are far too complex for small institutes with limited technical staff.

2.1.4 Desktop Applications (VB.NET / Access)

Desktop-based applications built with Visual Basic .NET and Microsoft Access are widely used in college projects. However, they are Windows-only, require local installation, and cannot be accessed remotely — making them unsuitable for modern institutional requirements.

2.2 Proposed System

The proposed system overcomes the above limitations through modern web technologies:

- **Web-based:** Accessible from any browser without installation.
- **Cloud Database:** Neon PostgreSQL ensures data availability, backups, and scalability.
- **Secure Auth:** NextAuth v5 with JWT and bcrypt password hashing.
- **Modern UI:** shadcn/ui with Tailwind CSS — clean, responsive, professional.
- **Real-time Search:** Instant filtering across all modules.
- **REST API:** Clean separation of frontend and backend, easy to extend.

2.3 Comparison of Existing and Proposed System

Table 2.1: *Comparison of Existing and Proposed Systems*

Feature	Manual	Excel	VB.NET App	Proposed System
Platform	Paper	Windows	Windows Only	Any Browser
Multi-user Access	No	Limited	No	Yes
Data Security	None	Password only	Basic	JWT + bcrypt
Search & Filter	Manual	Basic	Limited	Real-time
Report Generation	Manual	Manual	Basic	Automated
Cloud Storage	No	No	No	Yes (Neon DB)
Remote Access	No	No	No	Yes
Installation Required	No	Yes	Yes	No
Training Records	Register	Sheet	Basic	Full CRUD
Result Management	Register	Sheet	Basic	Full CRUD
Cost	Low	Low	Medium	Low (free tier)

Chapter 3

System Analysis and Requirements

3.1 Feasibility Study

3.1.1 Technical Feasibility

The system uses Next.js 16, TypeScript, Prisma ORM, and Neon PostgreSQL — all open-source, well-documented, and widely supported technologies. Development requires only a computer with Node.js installed. The cloud database provides a free tier sufficient for institutional use. The project is technically feasible.

3.1.2 Economic Feasibility

All technologies are free and open-source. Neon PostgreSQL offers a generous free tier with no licensing fees. Development was carried out by the student team as an academic project. The project is economically feasible at near-zero cost.

3.1.3 Operational Feasibility

The system's intuitive UI requires minimal training for administrative staff. Being web-based, it can be accessed from any device without software installation. The project is operationally feasible.

3.2 Functional Requirements

Table 3.1: *Functional Requirements*

Req. ID	Module	Requirement
FR-01	Auth	Admin must log in with username and password
FR-02	Auth	Unauthenticated users must be redirected to login
FR-03	Students	Admin can add, edit, delete, and view student records
FR-04	Students	Search students by name or roll number
FR-05	Students	View individual student profile with training & results
FR-06	Companies	Admin can add, edit, delete, and view company records
FR-07	Training	Admin can add, edit, delete, and view training records
FR-08	Training	Training status: Ongoing or Completed

Req. ID	Module	Requirement
FR-09	Results	Admin can add, edit, delete, and view result records
FR-10	Reports	Consolidated report view for all modules

3.3 Non-Functional Requirements

Table 3.2: *Non-Functional Requirements*

Req. ID	Category	Requirement
NFR-01	Security	All routes protected by JWT authentication
NFR-02	Security	Passwords stored as bcrypt hashes only
NFR-03	Performance	Page load under 3 seconds on standard connection
NFR-04	Usability	Clean, consistent UI across all modules
NFR-05	Reliability	All API errors return meaningful messages
NFR-06	Scalability	Database capable of handling growth in records
NFR-07	Compatibility	Works on Chrome, Firefox, Edge, and Safari
NFR-08	Maintainability	Modular, consistent code conventions

3.4 Use Case Description

Table 3.3: *Use Case Table – Admin*

Use Case	Actor	Description	Outcome
Login	Admin	Enter credentials to access system	Authenticated session created
Logout	Admin	End the session	Session cleared, redirect to login
Add Student	Admin	Fill student form and submit	Student created in database
Edit Student	Admin	Modify existing student details	Student record updated
Delete Student	Admin	Remove student from system	Record permanently deleted
Search	Admin	Type in search box to filter records	Filtered results in real time
View Profile	Admin	Click eye icon on student row	Individual profile page shown
Add Training	Admin	Select student, company, fill training details	Training record created
Add Result	Admin	Select student, enter result details	Result record created

Use Case	Actor	Description	Outcome
View Reports	Admin	Navigate to reports page	All module data shown together

Chapter 4

System Design

4.1 System Architecture

The system follows a **three-tier architecture** — Presentation, Application, and Data layers.

- **Presentation Layer:** React components (Next.js App Router) with shadcn/ui and Tailwind CSS handle all user interactions.
- **Application Layer:** Next.js API Routes implement the REST API. NextAuth v5 manages authentication. The proxy file enforces route protection at the edge.
- **Data Layer:** PostgreSQL on Neon DB, accessed via Prisma ORM with the Neon HTTP adapter for serverless-compatible queries.

Figure 4.1: *System Architecture – Three-Tier Web Application*

```
┌─────────────────────────────────────────────────────────┐
│ PRESENTATION LAYER (Browser) │
│ Next.js React · shadcn/ui Components · Tailwind CSS │
└────────────────────────┬────────────────────────────────┘
                         │ HTTP / HTTPS Requests
┌────────────────────────▼────────────────────────────────┐
│ APPLICATION LAYER (Next.js Server) │
│ REST API Routes · NextAuth v5 JWT · Prisma ORM │
└────────────────────────┬────────────────────────────────┘
                         │ Neon HTTP Adapter (SSL)
┌────────────────────────▼────────────────────────────────┐
│ DATA LAYER (Neon Serverless PostgreSQL) │
│ Admin · Student · Company · Training · Result Tables │
└─────────────────────────────────────────────────────────┘
```

4.2 Database Design

Table 4.1: *Student Table Schema*

Column	Type	Constraints	Description
id	INT	PK, AUTO INCREMENT	Unique identifier
rollNo	VARCHAR	UNIQUE, NOT NULL	Student roll number

Column	Type	Constraints	Description
name	VARCHAR	NOT NULL	Full name
email	VARCHAR	NOT NULL	Email address
phone	VARCHAR	NOT NULL	Contact number
address	VARCHAR	NOT NULL	Home address
course	VARCHAR	NOT NULL	Enrolled course
year	INT	NOT NULL	Current year (1/2/3)
createdAt	TIMESTAMP	DEFAULT NOW()	Creation timestamp
updatedAt	TIMESTAMP	AUTO UPDATE	Last update timestamp

Table 4.2: *Company Table Schema*

Column	Type	Constraints	Description
id	INT	PK, AUTO INCREMENT	Unique identifier
name	VARCHAR	UNIQUE, NOT NULL	Company name
industry	VARCHAR	NOT NULL	Industry sector
location	VARCHAR	NOT NULL	City / location
contactPerson	VARCHAR	NOT NULL	HR contact name
email	VARCHAR	NOT NULL	Contact email
phone	VARCHAR	NOT NULL	Contact phone

Table 4.3: *Training Table Schema*

Column	Type	Constraints
id	INT	PK, AUTO INCREMENT
studentId	INT	FK → Student.id, NOT NULL
companyId	INT	FK → Company.id, NOT NULL
projectName	VARCHAR	NOT NULL
startDate	TIMESTAMP	NOT NULL
endDate	TIMESTAMP	NOT NULL
status	VARCHAR	DEFAULT 'Ongoing'
description	TEXT	NOT NULL

Table 4.4: *Result Table Schema*

Column	Type	Constraints
id	INT	PK, AUTO INCREMENT
studentId	INT	FK → Student.id, NOT NULL
subject	VARCHAR	NOT NULL
marks	INT	NOT NULL

Column	Type	Constraints
maxMarks	INT	NOT NULL
grade	VARCHAR	NOT NULL
semester	INT	NOT NULL
year	INT	NOT NULL

4.3 Module Design

Each module has a dedicated page under `/dashboard/` and corresponding REST API routes under `/api/`:

- **Authentication:** `/login` page · `/api/auth` (NextAuth handler)
- **Dashboard:** `/dashboard` · `/api/dashboard` (summary counts)
- **Students:** `/dashboard/students` · `/api/students` + `/api/students/[id]`
- **Companies:** `/dashboard/companies` · `/api/companies` + `/api/companies/[id]`
- **Training:** `/dashboard/training` · `/api/training` + `/api/training/[id]`
- **Results:** `/dashboard/results` · `/api/results` + `/api/results/[id]`
- **Reports:** `/dashboard/reports` · reads from all existing API endpoints

Chapter 5

Implementation

5.1 Technology Stack

Table 5.1: *Technology Stack Details*

Layer	Technology	Version	Purpose
Framework	Next.js	16.2.10	Full-stack React framework with App Router and API Routes
Language	TypeScript	5.x	Statically typed JavaScript for safer, maintainable code
Styling	Tailwind CSS	4.x	Utility-first CSS for rapid, responsive UI development
UI Components	shadcn/ui	Latest	Accessible, composable component library
Icons	Lucide React	Latest	SVG icon set
ORM	Prisma	7.8.0	Type-safe database access layer
DB Adapter	@prisma/adapter-neon	7.8.0	Serverless-compatible Neon adapter for Prisma
Database	Neon PostgreSQL	15	Serverless cloud PostgreSQL database
Authentication	NextAuth v5	5.0.0-beta	JWT session management with Credentials provider
Password Hashing	bcryptjs	3.x	Secure password hashing

5.2 Module Implementation with Screenshots

5.2.1 Login Page

The login page is the entry point of the application presenting a card-based form. On successful authentication, a JWT session is created and the admin is redirected to the dashboard. Invalid credentials display an error message.

Login

Figure 5.1: *Login Page*

5.2.2 Dashboard

The dashboard displays four summary cards (total students, companies, training records, and results) and a table of recently added student records. The sidebar provides navigation to all modules.

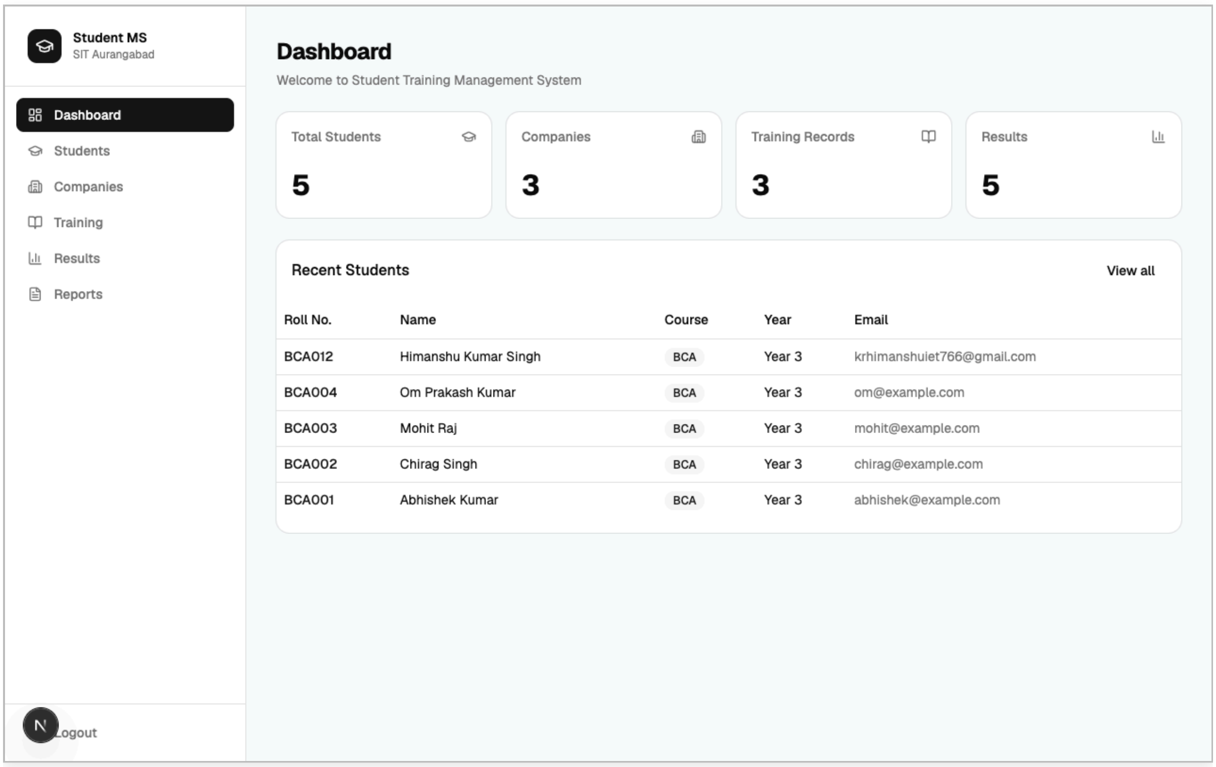

Figure 5.2: *Dashboard – Overview and Statistics*

5.2.3 Student Module

The student module provides a searchable table with roll number, name, email, phone, course, and year. A right-side drawer panel handles add/edit forms without page navigation. An eye icon navigates to the student's full profile page.

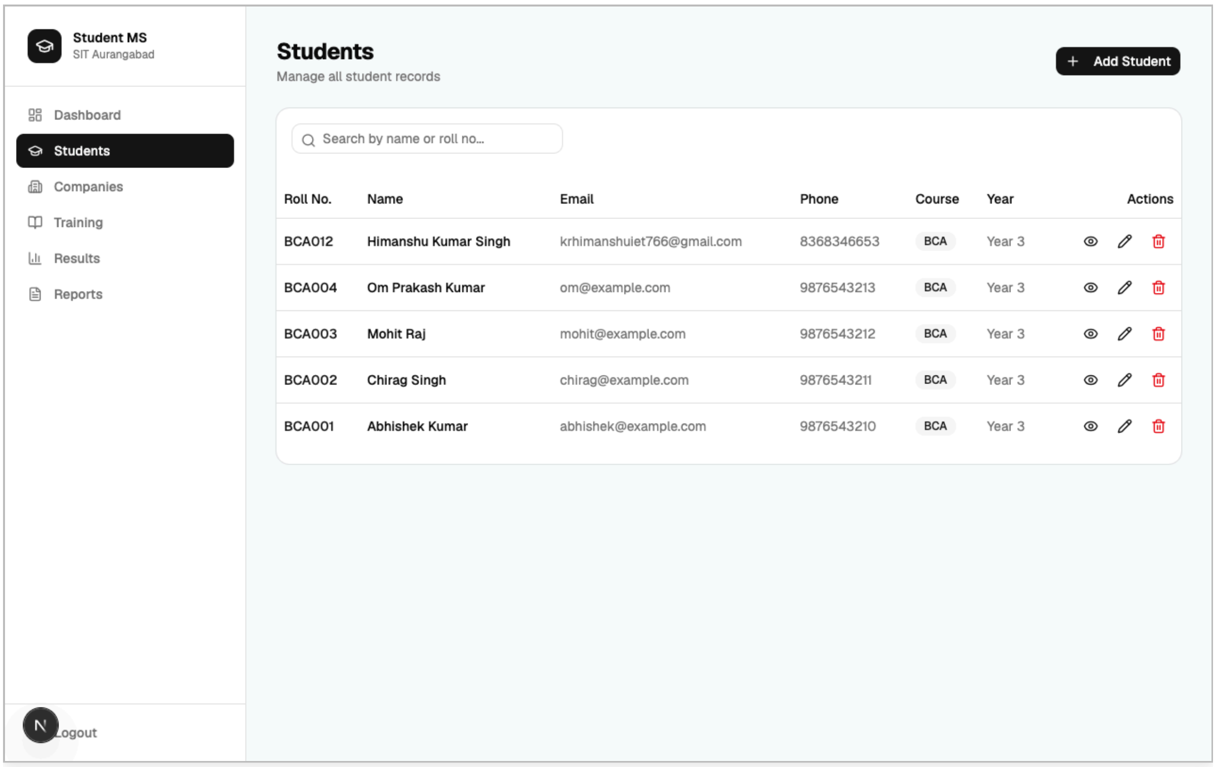

Figure 5.3: *Student Module – Records Management*

5.2.4 Company Module

The company module stores all organisations where students are placed for training. Each record holds the company name, industry, location, contact person, email, and phone. Companies can be searched by name.

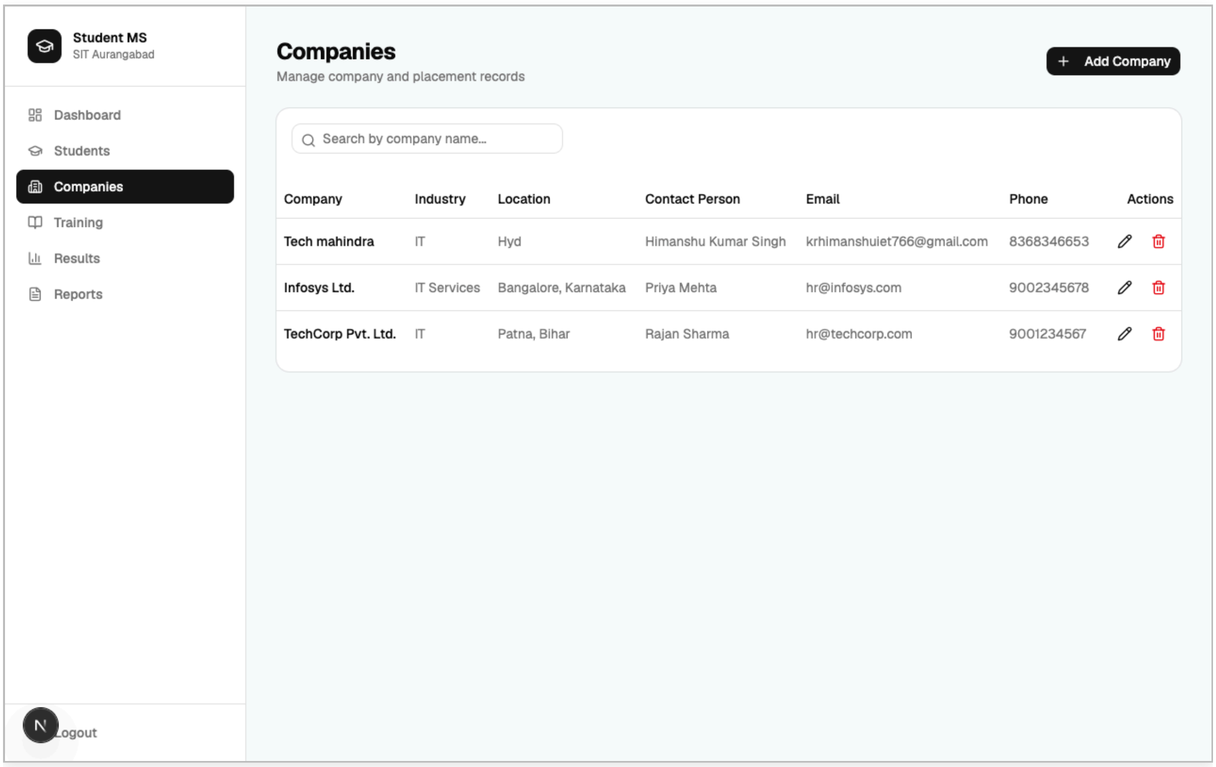

Figure 5.4: *Company Module – Placement Records*

5.2.5 Training Module

The training module tracks each student's internship. It records the student, company, project name, start/end dates, status (Ongoing/Completed), and a project description. Skeleton loaders appear during data fetch for better UX.

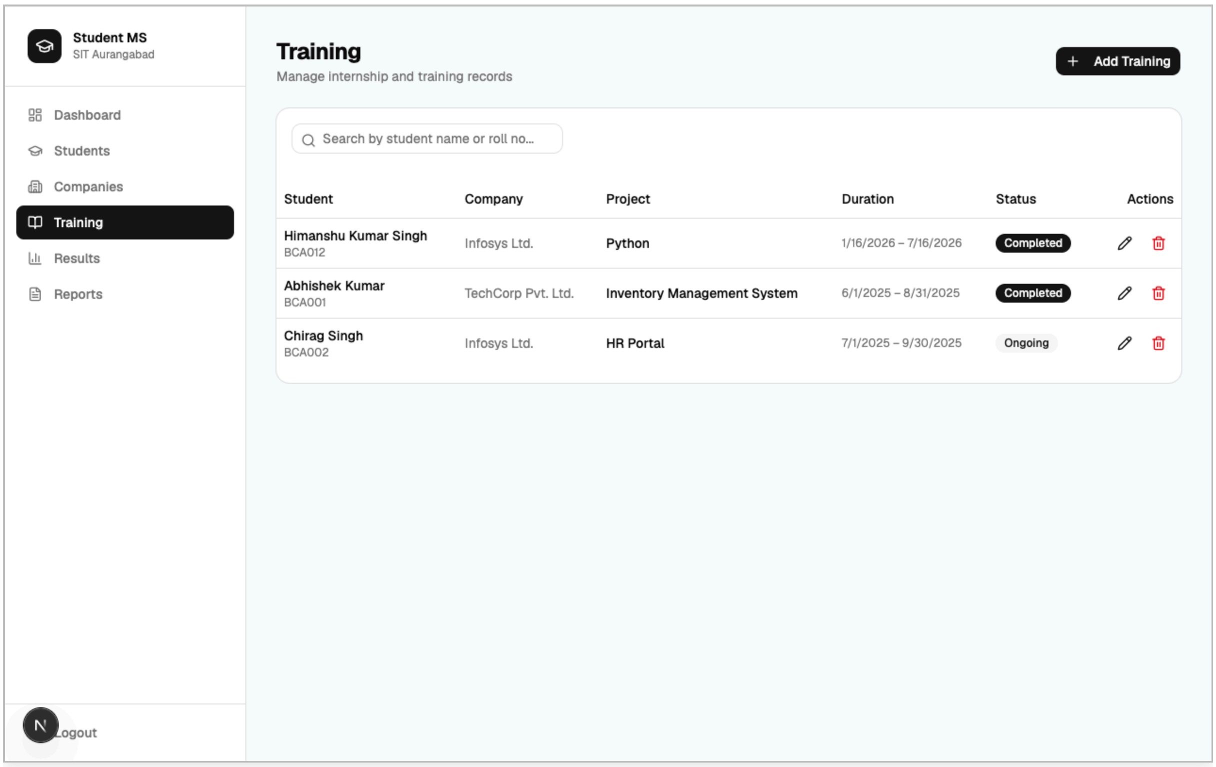

Figure 5.5: *Training Module – Internship Tracking*

5.2.6 Results Module

The results module records subject-wise academic performance. Each entry stores the student, subject, marks obtained, maximum marks, grade, semester, and year. Grades are displayed as colour-coded badges for quick visual scanning.

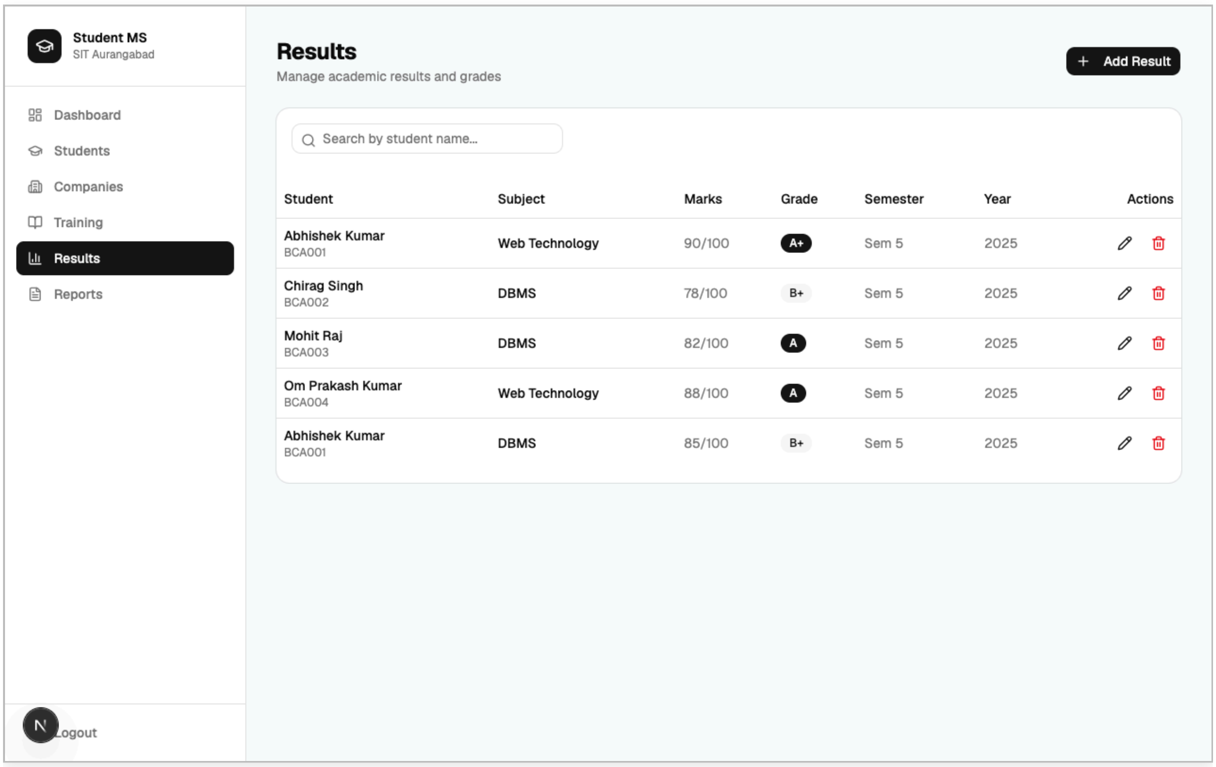

Figure 5.6: *Results Module – Academic Performance*

5.2.7 Reports Module

The reports module presents a consolidated, read-only view of all data — students, companies, training records, and results — on a single page, giving administrators a complete institutional overview without switching modules.

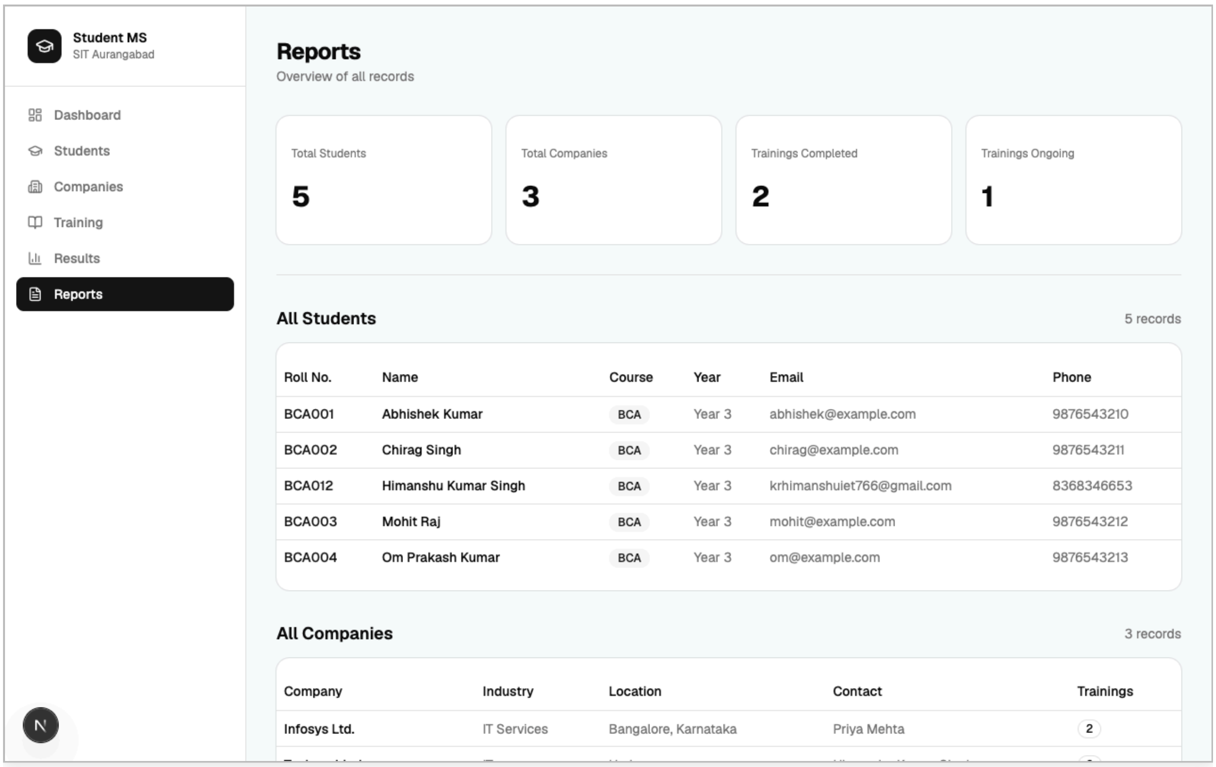

Figure 5.7: *Reports Module – Consolidated View*

Chapter 6

Testing

6.1 Testing Strategy

The following testing approaches were adopted:

- **Unit Testing:** Individual API routes tested with direct HTTP requests for valid and invalid inputs.
- **Integration Testing:** Frontend form submissions and backend API interaction verified end-to-end.
- **User Acceptance Testing (UAT):** All modules manually tested by the project team against specified requirements.
- **Boundary Testing:** Empty fields, invalid types, and extreme values tested to confirm proper validation.
- **Security Testing:** Unauthenticated access to protected routes confirmed to redirect to login.

6.2 Test Cases

Table 6.1: *Test Cases – Authentication Module*

TC#	Test Case	Input	Expected	Result
TC-01	Valid login	admin / admin123	Redirect to dashboard	✓ Pass
TC-02	Invalid password	admin / wrong	Error message shown	✓ Pass
TC-03	Empty username	"" / admin123	Form validation error	✓ Pass
TC-04	Access /dashboard without login	No session	Redirect to /login	✓ Pass
TC-05	Logout	Click logout	Session cleared, redirect	✓ Pass

Table 6.2: *Test Cases – Student Module*

TC#	Test Case	Input	Expected	Result
TC-06	Add student with valid data	All fields filled	Student record created	✓ Pass
TC-07	Duplicate roll number	Existing roll no.	Error: Roll number exists	✓ Pass
TC-08	Edit student details	Modified fields	Record updated in DB	✓ Pass

TC#	Test Case	Input	Expected	Result
TC-09	Delete student	Confirm delete	Record removed	✓ Pass
TC-10	Search by name	Partial name	Filtered results shown	✓ Pass

Table 6.3: Test Cases – Company & Training Modules

TC#	Module	Test Case	Expected	Result
TC-11	Company	Add company with valid data	Company record created	✓ Pass
TC-12	Company	Duplicate company name	Error: Company already exists	✓ Pass
TC-13	Company	Edit company details	Record updated	✓ Pass
TC-14	Company	Delete company	Company removed from list	✓ Pass
TC-15	Training	Add training with valid data	Training record created	✓ Pass
TC-16	Training	Submit without selecting student	Validation error shown	✓ Pass
TC-17	Training	Update status to Completed	Badge shows Completed	✓ Pass
TC-18	Training	Delete training record	Record removed	✓ Pass
TC-19	Results	Add result with valid data	Result record created	✓ Pass
TC-20	Results	Edit result grade	Grade updated in list	✓ Pass

6.2.1 Testing Summary

A total of 20 test cases were executed across Authentication, Student, Company, Training, and Results modules. All 20 test cases passed successfully, confirming that the system functions correctly as per the specified requirements. Error handling was verified with invalid and boundary inputs — in all cases the system responded with appropriate error messages rather than crashing or exposing internal details.

Chapter 7

Conclusion and Future Scope

7.1 Conclusion

The **Student Training Management System** has been successfully designed, developed, and tested as part of the Bachelor of Computer Application final year project at Sityog Institute of Technology, Aurangabad. The system fulfils all the objectives outlined in Chapter 1 and effectively addresses the problems identified in the problem statement.

The project demonstrates the practical application of modern full-stack web development technologies — Next.js 16, TypeScript, Prisma ORM, Neon PostgreSQL, NextAuth v5, and shadcn/ui. The resulting application is a robust, secure, and user-friendly platform that enables administrators to manage student records, company placements, training assignments, and academic results with ease.

Key achievements of this project include:

- A fully functional web-based system accessible from any modern browser.
- Secure authentication with JWT sessions and bcrypt password hashing.
- Complete CRUD operations across all four data modules.
- Real-time search and filtering for quick data retrieval.
- Skeleton loading states for improved user experience.
- Side drawer forms for smooth, in-page data entry.
- Individual student profile pages with complete history.
- A consolidated reports page for institutional overview.
- Cloud-hosted PostgreSQL database for persistent, scalable data storage.

The system successfully replaces manual paper-based and spreadsheet-based record keeping, providing significant improvements in efficiency, accuracy, and accessibility of institutional data management.

7.2 Future Scope

Several enhancements can be incorporated in future versions to make the system more comprehensive for larger institutional deployment:

- **Student Portal:** Allow students to log in and view their own training records, results, and profile.
- **Role-Based Access Control:** Multiple roles (Admin, Faculty, HOD) with different permission levels.
- **Email Notifications:** Automated emails to students when training or results are updated.
- **Attendance Management:** Daily attendance tracking with monthly summaries.
- **PDF Export:** Export individual student profiles, result sheets, and training certificates.
- **Mobile Application:** React Native companion app for on-the-go access.
- **Advanced Analytics:** Charts for placement trends, result distributions, and training statistics.
- **Multi-Campus Support:** Multiple departments and campuses within the same institution.
- **Placement Drive Module:** Manage drives, interview schedules, and offer letters.
- **SMS Notifications:** Alerts to students and parents about important events.

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